Bishal Dhungana

West Palm Beach, FL | beesal13dh@gmail.com | linkedin.com/in/dbishal | dbishal13.github.io | github.com/DBishal13

SUMMARY

Geospatial Data Scientist working across applied AI, cloud data platform engineering, and hazard-resilient infrastructure. As technical lead and supporting product owner on FPL's AI-based infrastructure inspection program, I independently built and deployed a computer vision model to production (AWS SageMaker, ECR), defined the ML scope and data pipelines behind a multi-vendor AI engagement, and built the GIS applications used for real-time disaster response. Background in field geomatics and peer-reviewed flood-hazard research. Open to Data Science, GIS/Geospatial Engineering, and AI/ML roles.

SKILLS

Machine Learning & Computer Vision: PyTorch, TensorFlow/Keras, Ultralytics, Roboflow, YOLO, RT-DETR, Detectron2, Faster/Mask R-CNN, EfficientDet, DETR, MMDetection, ResNet, XGBoost, OpenCV, ONNX

Generative AI & Agentic Systems: LangChain, LangGraph, RAG, Model Context Protocol, FastMCP, Vector Databases, Ollama, Chroma, Hugging Face Transformers

Data Engineering: Python, SQL, ETL/ELT Pipelines, DuckDB, Databricks, Data Lakes, Parquet

Geospatial: ArcGIS Pro / Enterprise / Online, Utility Network, PostGIS, DuckDB Spatial, GDAL, PDAL, GeoPandas, Shapely, PyProj, LiDAR / Point Cloud, Overture Maps, Cloud-Optimized GeoTIFF, Folium, FME

Cloud & DevOps: AWS Lambda, S3, SageMaker, ECR / ECS, AWS CDK, Docker, IAM, VPC / Network Security, CI/CD

Software Engineering: JavaScript / TypeScript, R, FastAPI, React, Streamlit

Field: GNSS / GPS Survey, UAV Survey (FAA Part 107), Differential GNSS, Geodatabase Design

EXPERIENCE

Data Scientist (Geospatial & AI) — Apr 2025 – Present

Florida Power & Light - Contract, West Palm Beach, FL

- Defined ML scope and technical specs as data scientist/supporting product owner for a multi-million-dollar AI vendor engagement; authored RFP technical requirements and served as technical SME through development and benchmarking.
- Independently built and deployed an overhead switch-state (open/closed) classification model, training it locally then migrating to AWS (ECR, SageMaker) for production inference.
- Contributed to testing and evaluating CV architectures (YOLO v8/v9/11, Faster/Mask R-CNN, EfficientDet, SSD, DETR/Deformable DETR, Detectron2, MMDetection, RT-DETR) against a precision/recall/F1 and image-scale evaluation methodology co-designed with the team, producing statistical evaluation reports for leadership; selected RT-DETR for transfer learning and standardized the approach for vendors.
- Built an API-based metadata pipeline with spatial (10m/30ft) and temporal (30-day) joins matching 350K+ condition reports to an 8M-image catalog, enriching ~2M images with condition metadata.
- Led the pipeline's move from ad hoc Python/GeoJSON scripts to an FME-to-PostGIS pipeline structuring 8M+ images and 65M+ associated annotations, with automatic time/version tracking via FME triggers. Designed the schema and cut 100+ metadata columns to ~50 with spatial indexing.
- Designed the annotation-storage schema for multiple model versions, and set up the taxonomy as a feature service with editor logging and concurrent multi-user updates, plus a separate read-only REST API for internal teams and vendors, replacing a fragile Excel-based process.
- Built a Streamlit app (later migrated to Power BI) with REST APIs summarizing dataset availability and taxonomy patterns for leadership.
- Acted as product owner for a dataset-creation plugin built with the enterprise IT team, defining filters, UI, and delivery logistics, and used it to deliver 300+ curated imagery datasets to vendors.
- Evaluated ~20 vendor bids with a GPT-based scoring agent, then audited 3 external ML vendors post-award for licensing compliance and supported resolving CUDA/GPU and batch-processing mismatches against AWS SageMaker.

- Co-developed an equipment detection and segmentation model (with a data scientist and software engineer) spanning 27-30 asset classes, evolving from object detection to segmentation.
- Designed an AWS CDK infrastructure-as-code framework with enterprise naming, tagging, and deployment patterns used across ML and application workloads.
- Built a data integration platform (Python, SQL, DuckDB) with a FastAPI/React console, and manage the PostGIS databases and AWS (Lambda, S3, SageMaker) architecture behind it.
- Built and deployed GIS applications (ArcGIS Enterprise, Experience Builder, Dashboards, Utility Network) for infrastructure monitoring and disaster response, including spatial web portals used in real time during Hurricane Milton for structural impacts, outages, and crew deployment.
- Built a storm-pilot deployment application in ArcGIS Experience Builder to manage internal and external UAV pilots, with real-time location tracking, multi-deployment coordination, time-of-deployment logging, and multi-level access for enterprise users.
- Used Roboflow for annotation and data preparation on both production CV models, including a 6x augmentation pipeline (flips, brightness, and other transforms), train/test/validation splits, and YOLO-format export.

Associate Geospatial Analyst Sep 2024 – Mar 2025

Florida Power & Light - Contract, Jupiter, FL

- Led QA/QC of LiDAR-derived power grid data for infrastructure resilience work, including pole hardening.
- Managed PostgreSQL/PostGIS databases, optimized Cloud-Optimized GeoTIFFs, and automated QA workflows in FME, ArcPy, and Model Builder to streamline data storage and validation.
- Built a FastAPI sidecar for heavier geoprocessing and format conversion.
- Built a suite of LiDAR-based transmission corridor tools: clearance and right-of-way mapping (GeoPandas, pykml, ArcPy); a point-cloud classification tool extracting pole height, length, top/bottom diameter, tilt, and communication attachment points; a CLI for vegetation-density classification (low/medium/high) supporting vegetation management at FPL's Turkey Point nuclear site; and a terrain-model generation CLI (GDAL, PDAL, GeoPandas) built against corridor LiDAR.

Research Assistant Jan 2023 – Aug 2024

Florida Atlantic University, Boca Raton, FL

- Conducted flood risk assessment for 3 Florida communities using FEMA hydrologic and hydraulic (H&H) modeling methods and Python automation, as part of FAU's Center for Water Resiliency and Risk Reduction (CWR3) Watershed Master Planning Initiative, a $1.7M FEMA-funded effort to build a statewide Watershed Master Plan template for Florida.
- Built an automated ArcGIS Pro geoprocessing pipeline, cutting processing time by 50%.
- Built a Python geocoding tool processing 3 million addresses, replacing a paid service; cut turnaround from 3 months to 1 week and saved $20K in licensing.

Geomatics Engineer Dec 2020 – Dec 2022

Freelance Consultant, Kathmandu, Nepal

- Led survey teams for municipal road, control, and construction surveys, including GNSS landmark and UAV topographic/transmission-line surveys.
- Designed municipal geodatabases (land use, soil, cadastral, resource maps) with topology and attribute rules.
- Served as GIS expert for Nepal's Survey Department land-use hand-over program, training 40+ officials in spatial analysis and database management.
- Digitized historical land records as part of a national land-use modernization effort, replacing manual paper-based archives.

EDUCATION

MSc, Geosciences — GIS Concentration — Florida Atlantic University · 2023–2024

BSc, Geomatics Engineering — Tribhuvan University, Nepal · 2016–2021

CERTIFICATIONS & TRAINING

- The Rise of the AI Data Engineer Bootcamp (2026)
- GIS Mapping & Spatial Analysis — U. Toronto

- FAA Part 107 — Remote Pilot Certificate
- Data Science with R — Johns Hopkins
- Machine Learning Specialization — DeepLearning.AI
- Remote Sensing Image Acquisition & Analysis — UNSW Canberra
- Deep Learning Specialization — DeepLearning.AI
- Introduction to Databases — Meta
- GIS Specialization — UC Davis

- **NASA ARSET Certifications (12):** Applied remote sensing training (2022-2024) in climate, disaster, and environmental risk modeling.
- **Esri Training Certifications (8):** GIS fundamentals, cartography, spatial data science, and imagery analysis.

PUBLICATIONS

Dhungana, B.; Liu, W. *Urban–Rural Exposure to Flood Hazard and Social Vulnerability in the Conterminous United States*. ISPRS Int. J. Geo-Inf. 2024, 13, 339.

Mandal, A.K.; Thapa, A.K.; Thapa Chhetri, M.; Dhungana, B.; Bloetscher, F.; Yong, Y.; Su, H. *PyWMP: A Unified Python Framework for Multi-Fidelity 1D, 2D, and Hybrid Event-Based Flood Modeling*. Manuscript in review, DOI pending; ongoing contributor to the open-source PyWMP package.

PROJECTS

SurgeExposure: A cloud-native pipeline overlaying NOAA storm-surge and flood-inundation data against Overture Maps building footprints, scoring per-asset flood exposure across 8 coastal regions — with a map and API. *(Python, DuckDB Spatial, FastAPI, PostGIS, AWS)* Demo | Repo

fflood-nep: A reproducible flood-response pipeline for Nepal's August 2026 Rasuwa/Bhote Koshi flash flood — runs Sentinel-1 SAR change detection for flood extent, scores building/road/facility exposure by municipality, and tracks a separate InSAR pipeline for landslide/avalanche source detection. Built and shipped inside the event's first week. *(Python, Sentinel-1 SAR, STAC, GDAL, InSAR)* Demo | Repo

wildfire-copilot: A LangGraph agent for utility wildfire risk — combines a trained XGBoost susceptibility model with live NASA/NWS hazard feeds and RAG over real wildfire-mitigation filings to answer plain-English risk questions and prioritize inspections. *(LangGraph, XGBoost, RAG, FastAPI)* Demo | Repo

geomlint: A CI-first data-quality linter for vector geospatial files — catches invalid geometry, CRS confusion, and topology errors that mainstream data-observability tools skip. No GDAL required; pip install and it runs anywhere pip does. *(Python, Shapely, PyProj, PyPI, CLI)* Docs | Repo

Geospatial Data Copilot: Ask a plain-English question about San Francisco's utility infrastructure — a local LLM agent (LangGraph ReAct, Ollama) translates it to spatial SQL, runs it, and answers with a summary and interactive map. No API keys. *(LangGraph, Ollama, DuckDB Spatial, Chroma)* Demo | Repo

HazardMCP: An MCP server unifying FEMA, USGS, NOAA, NWS, and NASA hazard data behind a single AI-agent interface — one-call location risk summaries and map rendering, compatible with Claude Desktop. *(Python, FastMCP, Folium, Docker)* Repo

Road-Centerline: A published Python package that extracts road centerlines from polygon geometries — worldwide, in any input CRS, in any format GeoPandas can read. CLI and Python API, computed via medial-axis skeletonization. *(Python, GeoPandas, pygeoops, PyProj, PyPI)* Repo

3DEP LiDAR Terrain Builder: Generates DEM, DSM, hillshade, slope, aspect, and contour products from USGS 3DEP lidar. Draw an area of interest in the browser; it generates a configured local script — no data ever uploads. *(PDAL, GDAL, RichDEM, AWS EPT/S3)* Demo | Repo

GeoCoder: Converts a spreadsheet of US street addresses into geocoded points and building footprint polygons using free Nominatim and Overpass APIs — as a no-install web app or a scriptable Jupyter notebook. *(GeoPandas, Nominatim / Overpass, Static Web App)* Demo | Repo

GTFS Transit Platform: Data-engineering and analytics platform for GTFS transit feeds — headways, stop coverage, and spacing anomalies, explored in an interactive map app. Demo | Repo

3d-export: 3D-style land cover export from Sentinel-2 tiles, clipped to any country and automated via GitHub Actions — a modernized take on the rayshader R workflow. Demo | Repo

RomanNepaliAI: Converts romanized Nepali to Devanagari live, client-side, plus a CLI that batch-translates SRT subtitles through pluggable backends. Demo | Repo

Conversational AI Assistant: LangChain RAG chatbot with a history-aware retriever for multi-turn, context-grounded conversation, citing sources. Runs locally via Ollama. Repo

compound-risk-pbc: Client-side visualization tool for compound risk scoring — interactive charts over probability-based combination models. Demo | Repo

fv-gwr: Analysis code behind a peer-reviewed study on urban–rural flood hazard exposure and social vulnerability across the U.S. Repo

Air-Pollution-in-Florida: Dashboard visualizing daily mean PM2.5 across Florida with an automated update pipeline. Repo

CURRENTLY BUILDING

groundtruth: An agentic tool-execution layer that wraps every LLM-driven GIS operation in deterministic CRS resolution, geometry validation, and topology checks.

vecinity: A vector index that spatially partitions as it searches, instead of bolting H3 cell IDs onto a generic ANN index as a metadata filter.

fv-gmgwr: GPU-accelerated Multi-scale Geographically Weighted Regression (MGWR) in PyTorch, extending the modeling approach behind my peer-reviewed flood-exposure paper (Dhungana & Liu, 2024).